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**ZU/WI/25/EXM/6**

**UNIVERSITY EXAMINATIONS 2025/2026**

**ORDINARY (ONLINE) EXAMINATION FOR DIPLOMA IN COMPUTER SCIENCE (DCS) AND DIPLOMA IN SOFTWARE ENGINEERING (DSE)**

**DMT 211 : PROBABILITY AND STATISTICS**

**DATE: APRIL, 2026** **Time: 2 Hours**

**INSTRUCTIONS**

1. Answer question **ONE (1)** and any other **TWO (2)** questions.
2. Scientific and non-programmable calculators may be used where necessary.
3. Use the provided statistical tables where applicable.

## QUESTION ONE (30 MARKS)

1. (a) Explain the meaning of the following terms as used in statistics.
   (i) Descriptive statistics and inferential statistics? ( 2 *marks*)
   (i) Null and alternative hypothesis. ( 2 *marks*)
   (ii) Type I and Type II errors. ( 2 *marks*)

   (b) Classify each of the following variables according to the appropriate level of measurement: Nominal, Ordinal, Interval,or Ratio. ( 4 *marks*)
   (i) Blood type (A, B, AB, O)
   (ii) Temperature in degrees Celsius
   (iii) Number of students in a class
   (iv) Ranking of runners in a race (1st, 2nd, 3rd, etc.)

   (c) Interviews for a job are carried out by two managers. Candidates are given a score by each of the two Managers X and Y to the eight candidates independently as shown in the table. Calculate the Spearman's Rank correlation coefficient for these data. ( 5 *marks*)

| Manager | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Manager X | 62 | 56 | 87 | 54 | 65 | 15 | 12 | 10 |
| Manager Y | 54 | 47 | 71 | 50 | 49 | 25 | 30 | 44 |

   (d) A restaurant is such that one of its dish gets ordered on average 4 times per day. What is the probability of having this dish ordered exactly 3 times tomorrow? ( 3 *marks*)

(e) The data below shows weights of 10 second year Journalism students:

$$60, 49, 57, 66, 79, 55, 43, 52, 62, 54, 60$$

Calculate;

(i) The Interquartile range (IQR) ( 3 *marks*)

(ii) $80^{th}$ percentile ( 2 *marks*)

(f) The table below gives a probability distribution of a discrete random variable $X$.

| $X$ | 40 | 80 | 120 | 150 | 200 |
|---|---|---|---|---|---|
| $P(X = x)$ | $a$ | 0.20 | 0.23 | $b$ | 0.15 |

Given that $P(X < 150) = 0.6$, find:

(i) the value of $a$ and $b$ ( 2 *marks*)

(ii) Expected value $\mathbb{E}[X]$ standard deviation of $X$. ( 5 *marks*)

## QUESTION TWO (15 MARKS)

2. (a) Use the frequency distribution data below to answer the questions that follow. The data represents the execution times (in milliseconds) of different algorithms.

| **Execution Time (ms)** | 5-14 | 15-24 | 25-34 | 35-44 | 45-54 | 55-64 |
|---|---|---|---|---|---|---|
| **Frequency (f)** | 4 | 6 | 5 | 3 | 2 | 1 |

(i) Find the mean execution time. ( 3 *marks*)

(ii) Find the standard deviation. ( 3 *marks*)

(iii) Determine the median. ( 4 *marks*)

(b) In a server rack, there are 3 Linux servers, 3 Windows servers and 4 macOS servers. If two servers are selected randomly for maintenance without replacement, determine the probability of selecting

(i) Two Windows servers ( 2 *marks*)

(ii) Servers of the same operating system ( 3 *marks*)

## QUESTION THREE (15 MARKS)

3. (a) In a coding competition, the probability of solving a difficult problem correctly is 0.25. Calculate the probability that in 15 problems a contestant solves a difficult problem correctly;

(i) Exactly 6 times ( 2 *marks*)

(ii) At least once ( 3 *marks*)

(b) Differentiate between a discrete random variable and a continuous random variable and give an example in each case ( 3 *marks*)

(c) An IT college has two classes, day class and evening class. From a survey conducted by the head of academics in the college the following results were obtained:

| | **Classes** | |
|---|---|---|
| | **Day** | **Evening** |
| Number of Students ($n_i$) | 13 | 15 |
| Average test marks ($\bar{x}_i$) | 45 | 55 |
| Standard deviation ($s_i$) | 4 | 5 |

**Required:** Determine whether there is any significant difference in the average test mark between the two classes at 5% level of significance. ( 7 *marks*)

## QUESTION FOUR (15 MARKS)

4. (a) An important application of regression analysis in accounting is in the estimation of cost. By collecting data on volume and cost and using the least squares method to develop an estimated regression equation relating volume and cost, an accountant can estimate the cost associated with a particular manufacturing volume. Consider the following sample of production volumes and total cost data for a manufacturing operation.

| **Production Volume(units)(X)** | 4 | 5 | 6 | 7 | 9 | 10 |
|---|---|---|---|---|---|---|
| **Total cost (Sh.000)(Y)** | 40 | 50 | 54 | 59 | 64 | 70 |

**Required**

(i) Draw a scatter diagram for the data. ( 3 *marks*)

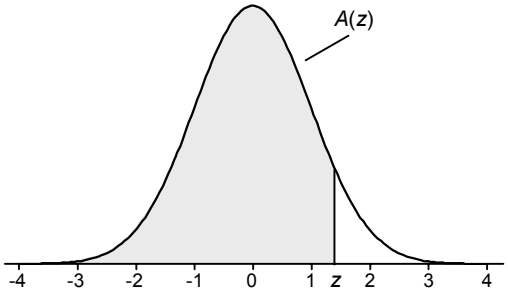
(ii) Estimate the simple linear regression relationship between total cost share and production volume: $Y = \alpha + \beta X$. ( 7 *marks*)

(iii) Estimate the total cost when $12$ units are produced. ( 1 *marks*)

(b) Probability sampling involves a mathematical chance of selecting the respondent. Explain any **two** probability sampling techniques. ( 4 *marks*)

## TABLE A.1

### Cumulative Standardized Normal Distribution




$A(z)$ is the integral of the standardized normal distribution from $-\infty$ to $z$ (in other words, the area under the curve to the left of $z$). It gives the probability of a normal random variable not being more than $z$ standard deviations above its mean. Values of $z$ of particular importance:

| $z$ | $A(z)$ | |
|---|---|---|
| 1.645 | 0.9500 | Lower limit of right 5% tail |
| 1.960 | 0.9750 | Lower limit of right 2.5% tail |
| 2.326 | 0.9900 | Lower limit of right 1% tail |
| 2.576 | 0.9950 | Lower limit of right 0.5% tail |
| 3.090 | 0.9990 | Lower limit of right 0.1% tail |
| 3.291 | 0.9995 | Lower limit of right 0.05% tail |

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |
| 3.5 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 |
| 3.6 | 0.9998 | 0.9998 | 0.9999 | | | | | | | |

## TABLE A.2

### *t* Distribution: Critical Values of *t*

| | | *Significance level* | | | | | |
|---|---|---|---|---|---|---|---|
| *Degrees of freedom* | *Two-tailed test*: *One-tailed test*: | 10% 5% | 5% 2.5% | 2% 1% | 1% 0.5% | 0.2% 0.1% | 0.1% 0.05% |
| **1** | | 6.314 | 12.706 | 31.821 | 63.657 | 318.309 | 636.619 |
| **2** | | 2.920 | 4.303 | 6.965 | 9.925 | 22.327 | 31.599 |
| **3** | | 2.353 | 3.182 | 4.541 | 5.841 | 10.215 | 12.924 |
| **4** | | 2.132 | 2.776 | 3.747 | 4.604 | 7.173 | 8.610 |
| **5** | | 2.015 | 2.571 | 3.365 | 4.032 | 5.893 | 6.869 |
| **6** | | 1.943 | 2.447 | 3.143 | 3.707 | 5.208 | 5.959 |
| **7** | | 1.894 | 2.365 | 2.998 | 3.499 | 4.785 | 5.408 |
| **8** | | 1.860 | 2.306 | 2.896 | 3.355 | 4.501 | 5.041 |
| **9** | | 1.833 | 2.262 | 2.821 | 3.250 | 4.297 | 4.781 |
| **10** | | 1.812 | 2.228 | 2.764 | 3.169 | 4.144 | 4.587 |
| **11** | | 1.796 | 2.201 | 2.718 | 3.106 | 4.025 | 4.437 |
| **12** | | 1.782 | 2.179 | 2.681 | 3.055 | 3.930 | 4.318 |
| **13** | | 1.771 | 2.160 | 2.650 | 3.012 | 3.852 | 4.221 |
| **14** | | 1.761 | 2.145 | 2.624 | 2.977 | 3.787 | 4.140 |
| **15** | | 1.753 | 2.131 | 2.602 | 2.947 | 3.733 | 4.073 |
| **16** | | 1.746 | 2.120 | 2.583 | 2.921 | 3.686 | 4.015 |
| **17** | | 1.740 | 2.110 | 2.567 | 2.898 | 3.646 | 3.965 |
| **18** | | 1.734 | 2.101 | 2.552 | 2.878 | 3.610 | 3.922 |
| **19** | | 1.729 | 2.093 | 2.539 | 2.861 | 3.579 | 3.883 |
| **20** | | 1.725 | 2.086 | 2.528 | 2.845 | 3.552 | 3.850 |
| **21** | | 1.721 | 2.080 | 2.518 | 2.831 | 3.527 | 3.819 |
| **22** | | 1.717 | 2.074 | 2.508 | 2.819 | 3.505 | 3.792 |
| **23** | | 1.714 | 2.069 | 2.500 | 2.807 | 3.485 | 3.768 |
| **24** | | 1.711 | 2.064 | 2.492 | 2.797 | 3.467 | 3.745 |
| **25** | | 1.708 | 2.060 | 2.485 | 2.787 | 3.450 | 3.725 |
| **26** | | 1.706 | 2.056 | 2.479 | 2.779 | 3.435 | 3.707 |
| **27** | | 1.703 | 2.052 | 2.473 | 2.771 | 3.421 | 3.690 |
| **28** | | 1.701 | 2.048 | 2.467 | 2.763 | 3.408 | 3.674 |
| **29** | | 1.699 | 2.045 | 2.462 | 2.756 | 3.396 | 3.659 |
| **30** | | 1.697 | 2.042 | 2.457 | 2.750 | 3.385 | 3.646 |
| **32** | | 1.694 | 2.037 | 2.449 | 2.738 | 3.365 | 3.622 |
| **34** | | 1.691 | 2.032 | 2.441 | 2.728 | 3.348 | 3.601 |
| **36** | | 1.688 | 2.028 | 2.434 | 2.719 | 3.333 | 3.582 |
| **38** | | 1.686 | 2.024 | 2.429 | 2.712 | 3.319 | 3.566 |
| **40** | | 1.684 | 2.021 | 2.423 | 2.704 | 3.307 | 3.551 |
| **42** | | 1.682 | 2.018 | 2.418 | 2.698 | 3.296 | 3.538 |
| **44** | | 1.680 | 2.015 | 2.414 | 2.692 | 3.286 | 3.526 |
| **46** | | 1.679 | 2.013 | 2.410 | 2.687 | 3.277 | 3.515 |
| **48** | | 1.677 | 2.011 | 2.407 | 2.682 | 3.269 | 3.505 |
| **50** | | 1.676 | 2.009 | 2.403 | 2.678 | 3.261 | 3.496 |
| **60** | | 1.671 | 2.000 | 2.390 | 2.660 | 3.232 | 3.460 |
| **70** | | 1.667 | 1.994 | 2.381 | 2.648 | 3.211 | 3.435 |
| **80** | | 1.664 | 1.990 | 2.374 | 2.639 | 3.195 | 3.416 |
| **90** | | 1.662 | 1.987 | 2.368 | 2.632 | 3.183 | 3.402 |
| **100** | | 1.660 | 1.984 | 2.364 | 2.626 | 3.174 | 3.390 |
| **120** | | 1.658 | 1.980 | 2.358 | 2.617 | 3.160 | 3.373 |
| **150** | | 1.655 | 1.976 | 2.351 | 2.609 | 3.145 | 3.357 |
| **200** | | 1.653 | 1.972 | 2.345 | 2.601 | 3.131 | 3.340 |
| **300** | | 1.650 | 1.968 | 2.339 | 2.592 | 3.118 | 3.323 |
| **400** | | 1.649 | 1.966 | 2.336 | 2.588 | 3.111 | 3.315 |
| **500** | | 1.648 | 1.965 | 2.334 | 2.586 | 3.107 | 3.310 |
| **600** | | 1.647 | 1.964 | 2.333 | 2.584 | 3.104 | 3.307 |
| **∞** | | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 |